B5GEMINI: Digital Twin Network for 5G and Beyond

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***Abstract*— Digital Twin Network (DTN) is a new technology that builds on the concept of Digital Twins (DT) to create a virtual representation of the physical objects of a telecommunications network. DTN bridges physical and virtual spaces to enable coordination and synchronization of physical parts while eliminating the need to directly interact with them. In this work, we present B5GEMINI a DTN for 5G and beyond networks that makes an extensive use of artificial intelligence (AI). First, we present the infrastructural and architectural components that support B5GEMINI. Next, we explore five paradigmatic use cases where AI can leverage B5GEMINI for building new AI-powered applications. Finally, we identify the main components of the AI ecosystem of B5GEMINI.**

***Index Terms*—digital twin, digital twin networks, artificial intelligence, machine learning, telecommunications**

I. INTRODUCTION

A DT is a virtual model that resemble the characteristics and behavior of a physical asset or system, modeling its components and properties, as well as the interactions of the entity with the environment. A DT can be used to monitor and manage a physical asset or system in real-time or use it to recreate its behavior in predefined virtual scenarios in order to define a better industrial process or detect problems before the real implementation.

A Digital Twin Network (DTN) is a virtual representation of a telecommunications network that accurately models the devices, communication links, operating environment and applications running on the network. DTNs are a new way of designing and managing networks, in which multiple physical assets and their corresponding virtual twins are connected together to share information and collaborate to complete a set of tasks [1]. In DTNs, network operators in coordination with service providers and telecommunication equipment suppliers can validate new functionalities prior to their incorporation into the network or emulate existing situations in the network and analyze the effect of applying different policies in the network.

By leveraging network data, DTNs can be used to build a virtual representation of a network. The data should replicate the expected behavior of the network, for this reason, the most common source are live feeds from the real network, although data collected from testbeds or simulations that reflect the intended behavior can also be used.

Although DTNs are still in their nascent stage, it is envisioned that they will have a pervasive impact on telecommunication industry, and they are pictured to be a keystone technology for the fourth industrial revolution. The pursue of this interest has led to an accelerated research effort in the last years, mainly because they can be used to thoroughly dissect the response of a network under different conditions and test new networking designs without compromising the safety of the physical network. Additionally, by creating a DTN, real time optimization of its corresponding physical equivalent is also possible. Furthermore, DTNs can help secure traditional networks by enabling fast identification and isolation of network failures to quickly respond to security threats.

In recent years, the use of artificial intelligence (AI) for the creation and operation of DTNs has begun to be explored. AI methods can be used to improve the accuracy of DTN simulations and to enable new services that ultimately enhance the user experience such as self-healing networks, autonomous network management, and security monitoring. In addition, AI methods can also be used to automatically learn and adapt to changes in the real assets. This can help ensure that the DTN remains accurate over time.

Machine Learning (ML) techniques can be used to extract insight from network data and construct the corresponding DTN of the network [2]. In this context, a DTN can be interpreted as a black-box ML model of a network that can be parameterized in multiple ways to recreate different network scenarios (e.g., different topologies, traffic volume, dynamic placement policies), by adjusting the parameter values it is possible to observe the resulting changes in the DTN in real time and evaluate the situation using different metrics defined according to the operator's needs. More interestingly, the above is possible without disrupting the real network or incurring the construction of costly test environments.

In this work, we make the case of ML methods as a key enabler for the construction of DTN that model complex 5G and beyond networks. We present our proposed architecture to facilitate the deployment of DTN for smart applications, focusing on the relevant infrastructural details and AI components that support its construction. In addition, we discuss the innovative use cases of AI methods applied to the creation or operation of DTNs to meet the control and monitoring requirements of the telecommunication industry. We also outline several future research directions and key challenges to facilitate DTN adoption and improve its utility.

The increasing excitement generated by the DTN concept in the last years has raised interest in the IETF standardization group. The NMRG group has started a draft which proposes a reference framework for DTN and their possible applications [4]. Among these applications, ML model training and validation is explicitly mentioned. It should also be emphasized that one of the main concepts in this draft is the recommendation to have an exchange of information between

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the real physical network and the DT to obtain maximum benefit.

In order to achieve the goals proposed by the IETF draft, we present B5GEMINI is a research project, whose objective is to deploy a 5G/6G Digital Twin Network (DTN) support platform capable of building a specific network configuration and generating the necessary traffic to be subsequently used for different analysis activities focused on advanced scenarios such as cybersecurity and network management, ranging from resource optimization to security enforcement. Due to the inherent complexity of 5G/6G models, this project aims to make and extensive use of AI to provide them with self-learning capabilities.

B5GEMINI will provide a virtualized representation of the network meant to analyze, diagnose, emulate, and control the physical network. The virtual system will start with a simple form of the real system and, using AI self-learning capabilities and data updating, evolve gradually to a more realistic image. With the support of AI, complex data sources and telemetry (e.g., IoT sensors, network data) will be fed into sophisticated data interpretation processes to facilitate the replica of 5G network components.

B5GEMINI addresses two complementary objectives: the development of a DTN infrastructure (B5GEMINI-INFRA subproject) and the design and integration of advanced AI components and use cases in the DTN for realistic experimentation (B5GEMINI-AIUC subproject).

The developed system will support a virtual image of physical network facilities (emulation) and will allow to deploy complex network scenarios in a controlled way, launching clients and servers and collecting the traffic generated by them even if they interact with clients and servers outside the DTNs, typically on the wider Internet.

DT technology applied to 5G and 6G networks in B5GEMINI is composed of the physical twin, the DTN, data flows, services, and connections between both twins. From this point of view, using a bidirectional data flow between both worlds, the DT is capable of continuously adapting to operational changes based on real-time data and information coming from the physical twin, being able, among other things, to predict the future state of the physical twin.

Based on the interconnection of this data flow, two concepts with differentiated characteristics are established [1]:

1. Digital Twin (DT): Treated as an intelligent system in charge of modeling all the characteristics of a specific physical component. In the case of DT, the communication performs a one-to-one mapping between the physical twin and its DT, allowing bidirectional feedback between them.

2. DTN (Digital Twin Network): Treated as an intelligent system made up of multiple individual DT systems, being able to model complex interactions. In this case, the data flow follows a many-to-many method, interconnecting all the DTs with each other within the DTN. In this way the exchange of information between all the DTs included in the DTN is facilitated, achieving more precise network status detections, enriching real-time analysis, and improving decision-making on physical twins.

B5GEMINI will make an extensive use of advanced AI mechanisms to deploy useful uses cases: (i) Design and development of ML components for cybersecurity and network management in real-time environments (e.g., deployment of realistic cryptomining or DNS over HTTP flood attack detectors), (ii) Intelligent support to DTN operations and AI-based management for orchestration and dynamic control of the deployment of instances and components of the DTN, (iii) Use of the DTN as a platform to perform distributed training and inference processes for ML and DL models using on-demand GPU virtualization, and (iv) ML as a Service to deploy off-the-shelf ML models in the DTN allocated resources to perform ML tasks with a high degree of efficiency avoiding costly training and testing tasks.

The remainder of this work is organized as follows: Section II reviews the literature on the application of AI methods in DTNs and presents the state-of-the-art in the field. Section III describes the advantages that DTNs can provide for infrastructure operation and depict the B5GEMINI architecture. Section IV details the main use cases that B5GEMINI will address, and the B5GEMINI AI components. Finally, Section VI summarizes the main findings of this work.

II. Related Work

We observed considerable research effort leveraging this technology towards the fulfillment of the expectations placed on 5G and beyond communications. The potential of edge and cloud computing platforms to manage DT-related data should be considered. In particular, edge computing allows leveraging the computing capabilities of distributed nodes at the edge of the network, allowing to reduce the bandwidth and dependency with the cloud gateway [5]. There is a large body of work focused on providing solutions to further minimize network and data processing latency between end users and edge servers to facilitate the adoption of intelligence-driven applications. In our research, we have observed a particular interest in looking for the most optimal ways to place and migrate DT in a network that combines these two paradigms [6], i.e., a DTN optimization task in which ML and DL techniques emerge as the most attractive solution. They propose a new approach based on DRL to find the optimal placement for DTs in order to reduce system latency. This algorithm considers both the placement strategy and the mobility of users. In addition, they suggest using transfer learning to migrate DTs to new users.

On the same line, Lu et al. proposed to combine DT with edge computing to efficiently connect IoT devices with CPS in an Industrial IoT scenario (IIoT) [6]. More specifically, they propose to model DTs of the physical devices present in the user layer and place them in the edge to allow their operation from the CPS to achieve a better optimization of the manufacturing process. For this purpose, they propose an asynchronous federated learning scheme to build the DT of those devices using ML techniques. This can be important for organizations that need to maintain the confidentiality of their data or have limited resources. Leveraging this concept, Lu et al. build their architecture based on this scheme to alleviate privacy concerns in DTN while reducing data transfer overhead.

To minimize energy consumption and reduce data transfers in a Mobile Edge Computing (MEC) scenario, Dong et al. propose an optimization of resource allocation and data offloading probabilities to the cloud, as well as an efficient mapping of the association between users and edge devices, by

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approximating these strategies using DL models in a supervised manner [7]. For this purpose, a DTN of the communication network is constructed and deployed on a central server, different configurations are evaluated in the DTN, and the best ones are used to train a DNN model in an off-line fashion.

Groshev et al. describes the key technologies that will support the deployment and operation of a DTN in an industrial setting, including edge and fog computing, and 5G network [8]. Additionally, they determine the role of AI methods to address some of the challenges posed in this scenario. More interestingly, they identify AI as a tool to optimally allocate and automatically scale computing resources for a DTN to satisfy key performance indicators (KPIs) such as latency and security requirements. In addition, they conduct an experimental evaluation as a proof-of-concept to predict the movement of a robotic arm using data collected from a DT to train a variety of ML models that are subsequently compared.

Zhao et al. propose to create a DTN of a Software-defined Vehicular Network (SDVN) as a simulation environment to evaluate the quality and reliability of ML and DL models before applying them in the physical network [9]. This simulations environment, which they call "Intelligent Digital Twin" (IDT), allows to assess whether the models remain robust to the intrinsic dynamic changes of a VN and retrain the models if needed.

In the telecommunications industry, a DTN can be used to monitor and operate complex communications networks. Network operators can use this technology to monitor and assess the impact of different 5G implementation strategies, while providing automated recommendations to improve network performance. Exploiting the data collected from the DTN, ML algorithms can be used to fine-tune these strategies to maximize utilization and optimize operational costs, thus serving as a valuable tool for network providers to deploy 5G network coverage faster and more efficiently.

From a more creative point of view, there have been several interesting proposals that attempt to determine other roles that ML can play within DTN. One of those interesting use case of ML in the operation of DTN is presented in [1]. As noted by the authors, the behavior of the virtual twin is expected not to disturb the physical space. To that end, ML techniques could be used to validate the next actions of the real system using data generated from the DTN model to prevent unwanted actions in the physical environment. In this scenario, which is also envisaged in [10], [2], ML would serve as an anomaly detection tool for network operators, enabling them to quickly anticipate abnormal situations and mitigate further damage.

Nguyen et al. anticipate 5G-based DTN technology as a city management service that can provide situational awareness and decision-making for city managers and first responders [2]. In particular, they present the case for DTN as a ML-powered tool for the prediction of infectious disease outbreaks. More specifically, they propose to create a DTN that represents the city infrastructure and the overlaying 5G network. With the increased bandwidth and low latency that 5G networks provide, it will be possible to collect and analyze data from a large number of sensors in real time to track people movement, activities, and interactions. Combining this information with past and current epidemiological data and using a combination of ML and conventional data analytic processes, it is possible to predict the most likely outbreak scenarios to occur. In this way, public health officials and other related professionals can be better prepared to develop strategies to respond to these situations more effectively. Extending this idea, the combination of 5G-based DT and ML can be applied to the development of more efficient and ambitious smart city applications, such as intelligent transportation, smart energy, and intelligent security.

ML and DL methods are expected to play an important role in DTN model building, as anticipated in [10], but, to this day, there is a need for further systematic research in this area to find the most suitable methods and models for different types of data and to determine how these models will interoperate with the rest of elements and contribute to the DTN.

III. B5GEMINI Architecture

The design proposal for a modular and scalable system for creation of DTN applied to 5G networks aims to model each of the elements of the 5G networks in the form of a DT, establishing a complete DTN model capable of accurately emulating the behavior of a 5G network in certain use cases, which will make use of the DTN to be able to apply task based on AI. Figure 2 shows the conceptual design of this complete system.

Next, each of the modules that make up the entire system will be reviewed.

A. System Input for DT Generation

In the phase of creation of the different DTs that will make up the DTN, two general system inputs are stablished, automatic entry and manual entry.

1) Automatic Entry: It is defined with the aim of obtaining an exact replica of the physical object to be modeled, the development of smart agents that can be deployed in the target network, in charge of collecting all the information necessary for the DT generation (topological information, hardware and software information, states, etc.). This information is introduced to the deployment module in JSON or XML format.

2) Manual Input: In the case of not having sufficient permissions to be able to deploy the intelligent agents described in the previous point, an option for loading these files manually by the operators is enabled, avoiding limiting the emulation capacity of the system.

B. Deployment Module

Once the initial phase of collecting information for the generation of the DTs is finished, the implementation module is used, which through the Terraform IaaS (Infrastructure as a Service) tool will oversee the implementation of the virtual infrastructure that will host the DTs. This module allows, through different providers, to deploy the infrastructure in any available cloud (AWS, Google Cloud, Azure, IBM, etc.) or in own virtualization infrastructures, using managers such as vSphere for VCenter (VMWare). The main advantage of this technology is the abstraction and non-dependence on specific API for each cloud, since once the Terraform deployment file is generated, it is easily portable to any other cloud that offers a provider compatible with it. Added to the advantage of deployment over any type of infrastructure is the resource management offered by Terraform, allowing the creation of multiple DTN deployments, isolated from each other, as well

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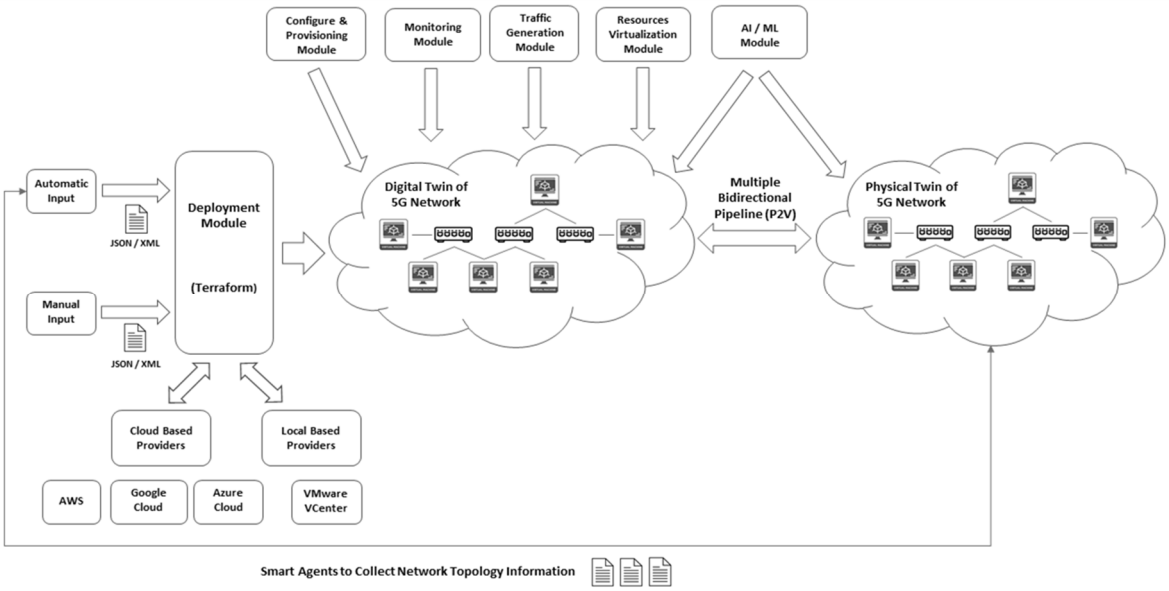


Fig. 1. B5GEMINI Architecture Diagram.

as their subsequent elimination, both at the full deployment level and for specific resources within a deployment. Finally, the deployment module makes use of a wide library of virtual machines and Docker containers, which will serve as the basis for the provisioning and configuration of the different DTs to be implemented.

C. Digital Twin Configuration and Provisioning Module

After completing the infrastructure deployment phase, the configuration and provisioning module is responsible, based on the information collected by the smart agents, for modeling each of the DT and configuring the necessary interconnections to fully simulate the target 5G network.

D. Network Monitoring Module

This module is responsible for monitoring the complete operation of all information exchanged within the DTN. Once all the DTs are provisioned and configured, the monitoring module will control the activation of port-mirroring functionalities within each one of the subnets present in the resulting DTN with the aim of being able to use this information traffic for the generation of datasets that allow improving the complete emulation of the DTN.

E. Traffic Generation and Injection Module

This module offers the option of enabling the traffic generation, based on different 5G traffic models over the DTN, and allowing the validation and analysis of network performance.

F. Bidirectional Pipeline

Once the DTN infrastructure has been deployed and provisioned, it is necessary to establish the different pipelines to be used in the complete system. Depending on the type of elements interconnected and their communication necessities, two different pipelines are distinguished: V2V (virtual to virtual) and P2V (physical to virtual). The former is performed within the DTN, interconnecting each of the DTs that compose it. These pipelines allow reflecting the behavior of the communication that takes place in the physical world, without the time limitation present in the physical world, allowing the communication of large data flows in reduced times, which facilitates the emulation task, accelerating the obtention of results. The second supervises the interconnection between the virtual world of the DTN and the physical world, allowing the continuous feedback loop between the DT and its physical counterpart, providing the system with continuous co-evolution and cooperation capabilities.

G. AI / ML Module to Drive Smart Actions

This module will provision the smart agents, which are AI / ML models that can perform a variety of tasks such as optimizing network policies (e.g., traffic routing, resource deployment strategies, etc.) or predicting network failures. For this purpose, this module will continuously monitor the DTN and act accordingly on it to perform some arbitrary task. These changes will then be replicated in the physical world through the P2V pipeline.

IV. AI Applications in B5GEMINI

The seminal work and corner stone of B5GEMINI is the Mouseworld Lab [11], a controlled environment set up in Telefónica I+D premises for running experiments that allow deploying complex network scenarios in a controlled way and generating realistic labelled data sets for training supervised ML components and validate supervised and unsupervised solutions. The Mouseworld Lab provides a way to launch clients and servers, collect the traffic generated by them even if they interact with clients and servers outside the Mouseworld on the Internet, and finally add labels to this traffic without operator intervention. This environment is deployed on an NFV-enabled architecture, under the management of an orchestrator (NFVO), extending an ETSI NFV MANO stack as necessary.

It is worth noting that AI and ML are included by design in the Mouseworld Lab to allow the deployment of realistic network management and cybersecurity use cases where ML could be seamlessly trained, tested, and deployed. Therefore, the Machine Learning Orchestration is part of the Digital Twin Entity Management and is composed of a Topology Generator

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and an Experiment Launcher. The first uses predefined templates, interacts with OSM (ETSI NFV SOL-005 interface) and Openstack (Glance Image service API) provisioning NSD (Network Service Descriptors) and VNFD (VNF descriptors) compatible with OSM, detailing the network functions, the links, and their day-1 configurations. The Experiment Launcher makes day-2 configurations and trigger the emulations functions for dataset generation, and it is described with OSM's ProxyCharms or with dedicated scripts. The current version uses a configuration file to define the statistical distribution of the traffic, the number of intervals in which the experiment is divided and the duration, and the type of service is emulating. Currently, there are defined two scenarios: crypto mining malware detection and DNS attacks.

Hereafter, we describe innovative AI applications that B5GEMINI, as a DTN platform, will allow to deploy.

A. DTN for Smart Applications

There is an increasing need to design realistic virtual network scenarios in which different types of traffic can be generated in a controlled way and ML models can be trained and validated. B5GEMINI provides these capabilities allowing the generation and capture of network traffic in a controlled manner to be used for training and testing ML components with realistic data. It should be noted that since the collected traffic is generated in a controlled environment, data labelling, which is required for training and testing supervised ML components, can be done without human intervention. This capability facilitates the labelling of data sets in the big data regime that are required to train and validate complex Deep Learning (DL) models. In addition, B5GEMINI replicates physical network elements and therefore, ML and DL components can be deployed and validated in real-time scenarios. Furthermore, using these approach, different problems and faulty behaviors that rarely occur during the real network operation can be easily replicated now without negatively affecting the network. In this way, this approach allows to train ML components for those situations in a more cost-effective and safe manner in order to ensure that they operate accordingly when they occur in real-world situations.

B. ML for DTN Deployment and Configuration

B5GEMINI ambition is to be able to use AI-based management for orchestration and dynamic control of the deployment of instances and components of the DTNs, which allow us to obtain the best results regarding the problem posed. The dynamism of a DTN allows to represent multiple realistic situations in a network: network connection topologies variability (e.g., ring or star architectures, redundant paths), location of specific functions (e.g., content caching, traffic filtering, hardware accelerators), network links capacity or technologies used, nodes and links status change (congestion, drop or capacity variation), etc. Taking advantage of the DTN software configuration flexibility, different problems can be simulated and changes in the DTN instantiation introduced. The gathered metrics and the data generated in the DTN can be used to train a ML model. Also, for some cases, such as Reinforcement Learning (RL) the ML model proposes changes in the DTN and once done the result can be evaluated. Some representative use cases in B5GEMINI are network path optimization for a service, network growth or reconfiguration based on traffic growth patterns, network problem root cause identification or optimal mitigation to a network attack.

C. Distributed Training and Inference on Demand

We propose to use the DTN as a platform to perform distributed training and inference processes for ML and DL models. The approach consists in virtualizing, instantiating, and allocating virtual GPUs within a DTN on demand to provision all the required resources to train and evaluate models. The proposed approach can facilitate federated learning schemes and provide increased capabilities for the next-generation edge computing paradigm. It should be noted that to this day the virtualization process relies heavily on proprietary technology (see the case of NVIDIA and its industrial-grade GPUs). We consider this situation as the major challenge that may hinder the adoption of this technology at scale, and, for this reason, we look forward the development of more open solutions to eventually materialize this concept.

D. DTN for ML as a Service

In addition, it will also be possible to deploy off-the-shelf ML models in the allocated resources that are prepared to perform a specific task with a high degree of efficiency. We propose to realize this idea providing a catalog of pre-trained models for a wide variety of tasks, the users may select one of these models depending on their requirements to perform some downstream task on the DTN. For instance, this approach will make it possible to deploy classification models at specific locations in the network (routers, terminals, etc.) to filter traffic coming from certain subnets where congestion has been detected according to certain rules. In this way, it will be possible to improve network performance while minimizing the use of resources. Additionally, once deployed, the models can be retrained regularly in a self-learning fashion to improve their overall performance and robustness over time.

E. DTN for Energy Consumption Optimization

The need for energy consumption has been increasing in communication networks as they have become more sophisticated and widespread over the years. To cope with the high energy demand, network providers must be able to manage their energy resources more efficiently. ML has proven to be a valuable tool to achieve this goal [3]. The same can be applied to a DTN scenario using ML methods to optimize the network's energy usage by predicting when and where energy usage is highest. In this way, the network can be reconfigured to reduce energy consumption in those specific areas. In addition, ML can be used to predict times of peak energy consumption so that the network can prepare to cope with increased demand.

Finally, we enumerate various AI components to be integrated in B5GEMINI to enable the deployment of smart applications at scale:

A. Synthetic Data Generation

The generation of synthetic network and user data using Generative Adversarial Networks (GANs) will avoid the privacy violations that can be incurred in a DTN when using data collected from real user and the lack of publicly available data for ML training and testing purposes [12], [13]. Moreover, GANs will make it possible to generate synthetic data of sufficient quality to completely replace real data, improving security and avoiding privacy leakage.

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B. Automatic Model Optimization

The development of novel Automated ML (AutoML) techniques in the context of DTNs will provide a mechanism to generate optimized AI and ML components that can be considered sustainable from an energy consumption point of view (Green-AI principles). For example, to generate compact ML-based attack detectors to be deployed in resource-limited edge nodes.

C. Novel Deep Learning Architectures

Graph Neural Networks (GNN) are neural network architectures able to process graphs as input to produce useful embeddings exploiting the concept of message passing between nodes in the graph [14]. We envision GNN as a key player in the construction of DTN for communication networks modelling as several studies have shown that GNNs are able to remain robust to changes in network topology, routing policies and traffic distribution [15].

D. Explainability

Another key aspect of networks operation is the necessity to understand the intentions and guarantees of the applied policies and the possibility to extract knowledge about failure cases when they occur. Moreover, ML model interpretability is of extreme significance to explain results obtained from the envisioned models and help to distinguish adversarial from legitimate inputs.

E. Intent-based Networking

Intent-based networking is an approach to network operation that focuses on user intent rather than network details. In an intent-based network (IBN), operators can define the desired objectives or outcomes and then rely on the network to automatically provision all the required resources and configure itself to achieve those objectives. Providing a component to describe in a declarative way the elements of a network would help to configure non-trivial DTN deployments.

VI. Conclusions

This article presents B5GEMINI, a DTN for B5G networks, and discusses the role of AI in the construction, deployment, and operation of it. To effectively enable all these ideas, we propose B5GEMINI as an approach for a modular and scalable DTN system over B5G networks that has the potential to serve as a platform for developing and deploying innovative intelligence-driven DTN applications. We describe the architecture of our system and present all the interconnected modules that comprise it, spanning from the infrastructure level to ML applications. We detail the use cases in which AI components can provide an added value to the DTN and the main components that will be part of the AI ecosystem in B5GEMINI.

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VIII. References

[1] Y. Wu, K. Zhang, and Y. Zhang, 'Digital Twin Networks: A Survey', *IEEE Internet of Things Journal*, vol. 8, no. 18, pp. 13789–13804, Sep. 2021, doi: 10.1109/JIOT.2021.3079510.

[2] H. X. Nguyen, R. Trestian, D. To, and M. Tatipamula, 'Digital Twin for 5G and Beyond', *IEEE Communications Magazine*, vol. 59, no. 2, pp. 10–15, Feb. 2021, doi: 10.1109/MCOM.001.2000343.

[3] D. Sesto-Castilla, E. Garcia-Villegas, G. Lyberopoulos, and E. Theodoropoulou, 'Use of Machine Learning for energy efficiency in present and future mobile networks', in *2019 IEEE Wireless Communications and Networking Conference (WCNC)*, Apr. 2019, pp. 1–6. doi: 10.1109/WCNC.2019.8885478.

[4] C. Zhou *et al.*, 'Digital Twin Network: Concepts and Reference Architecture', Internet Engineering Task Force, Internet Draft draft-zhou-nmrg-digitaltwin-network-concepts-06, Dec. 2021.

[5] Y. Ai, M. Peng, and K. Zhang, 'Edge computing technologies for Internet of Things: a primer', *Digital Communications and Networks*, vol. 4, no. 2, pp. 77–86, Apr. 2018, doi: 10.1016/j.dcan.2017.07.001.

[6] Y. Lu, S. Maharjan, and Y. Zhang, 'Adaptive Edge Association for Wireless Digital Twin Networks in 6G', *IEEE Internet of Things Journal*, vol. 8, no. 22, pp. 16219–16230, Nov. 2021, doi: 10.1109/JIOT.2021.3098508.

[7] R. Dong, C. She, W. Hardjawana, Y. Li, and B. Vucetic, 'Deep Learning for Hybrid 5G Services in Mobile Edge Computing Systems: Learn From a Digital Twin', *IEEE Transactions on Wireless Communications*, vol. 18, no. 10, pp. 4692–4707, Oct. 2019, doi: 10.1109/TWC.2019.2927312.

[8] M. Groshev, C. Guimarães, J. Martín-Pérez, and A. de la Oliva, 'Toward Intelligent Cyber-Physical Systems: Digital Twin Meets Artificial Intelligence', *IEEE Communications Magazine*, vol. 59, no. 8, pp. 14–20, Aug. 2021, doi: 10.1109/MCOM.001.2001237.

[9] L. Zhao, G. Han, Z. Li, and L. Shu, 'Intelligent Digital Twin-Based Software-Defined Vehicular Networks', *IEEE Network*, vol. 34, no. 5, pp. 178–184, Sep. 2020, doi: 10.1109/MNET.011.1900587.

[10] T. Alsboui, L. Qin, R. Hill, and H. Al-Aqrabi, 'Distributed Intelligence in the Internet of Things: Challenges and Opportunities', *Computer Science*, vol. 2, no. 4, Jul. 2021, doi: 10.1007/s42979-021-00677-7.

[11] A. Pastor, A. Mozo, D. R. Lopez, J. Folgueira, and A. Kapodistria, 'The Mouseworld, a security traffic analysis lab based on NFV/SDN', in *Proc. of the 13th International Conference on Availability, Reliability and Security*, New York, NY, USA, Aug. 2018, pp. 1–6. doi: 10.1145/3230833.3233283.

[12] A. Mozo, Á. González-Prieto, A. Pastor, S. Gómez-Canaval, and E. Talavera, 'Synthetic flow-based cryptomining attack generation through Generative Adversarial Networks', *Sci Rep*, vol. 12, no. 1, p. 2091, Dec. 2022, doi: 10.1038/s41598-022-06057-2.

[13] Á. González-Prieto, A. Mozo, S. Gómez-Canaval, and E. Talavera, 'Improving the quality of generative models through Smirnov transformation', *arXiv:2110.15914 [cs]*, Oct.

[14] Z. Wu, S. Pan, F. Chen, G. Long, C. Zhang, and P. S. Yu, 'A Comprehensive Survey on Graph Neural Networks', *IEEE Trans. Neural Netw. Learning Syst.*, vol. 32, no. 1, pp. 4–24, Jan. 2021, doi: 10.1109/TNNLS.2020.2978386.

[15] K. Rusek, J. Suárez-Varela, P. Almasan, P. Barlet-Ros, and A. Cabellos-Aparicio, 'RouteNet: Leveraging Graph Neural Networks for network modeling and optimization in SDN', *IEEE J. Select. Areas Commun.*, vol. 38, no. 10, pp. 2260–2270, Oct. 2020, doi: 10.1109/JSAC.2020.3000405.

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